

Raman Kumar

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Summary

Backend Developer with **1.5 years** of experience building production-ready microservices using Java, Spring Boot, and Python. Architected end-to-end backend systems, integrated **AI agents**, engineered secure authentication and authorization flows with gateway routing for distributed microservices. Contributor to the open-source community with **5+ merged PRs**.

Technical Skills

Languages: Java, Python (hands-on experience)

Frameworks & Libraries: Spring Boot, Spring Web, Spring Security, Spring Cloud, Spring Data, HATEOAS, AspectJ, Resilience4j

Cloud & Infrastructure: AWS, GCP, Utho, Runpod, Docker, Kubernetes, GitHub Actions, Kong Gateway

Databases: MySQL, MongoDB, H2, Redis

Version Control: Git, GitHub, GitLab, Bitbucket

Testing: JUnit, Mockito, Spring Test

APIs & Communication: REST, WebSocket, WebRTC

Build Tools: Maven, Gradle, Pip

IDEs: IntelliJ, VS Code, PyCharm

Work Experience

Backend Developer, Softnerve Technologies, Remote

Oct 2024 - Present

- Engineered a modular backend architecture spanning **6+ microservices** (Core Business Services, Security, Config Server, AI Agents, Containerized Browser Nodes, a custom Spring Boot Starter, and Kong Gateway).
- Architected a centralized security microservice implementing OAuth2-based JWT authentication, additionally integrating Google OAuth client login and two-factor authentication (**2FA**). Reduced token verification time by **30%**.
- Built a custom Spring Boot Starter dependency acting as a bridge between AI agents and EHR systems, providing annotation-based intent descriptors for service methods to enable AI execution, exposing APIs to serve schemas to the LLM and dispatch intent requests for method invocation, and securing all interactions using mutual TLS (**mTLS**).
- Integrated multi-tenant, per-user AI agents into backend workflows, supporting **2 agent modalities** (Playwright-based browser automation and a voice-based agent) with custom tool integrations, enabling concurrent execution for **~7 users per pod** through a custom WebSocket communication layer.
- Developed a Containerized Browser Nodes microservice to launch isolated browser instances inside Docker, exposing CDP for automation agents and VNC for user interaction, supporting **~10 concurrent browser sessions per pod**.
- Implemented real-time environment property refresh using Kafka Bus, supporting both push- and pull-based updates; automated configuration refreshes via Kubernetes CronJobs, reducing configuration update time by **90%**.
- Designed and maintained **30+ REST APIs** for core business services, authoring comprehensive unit and integration tests using JUnit and Mockito, achieving **95% test coverage**.
- Implemented and managed Kong Gateway using KIC for routing **10+ microservices**, improving API response time by **30%**. Developed a custom Lua-based Kong plugin integrated with the Security microservice to enforce JWT-based authentication and authorization, enabling per-API role-based access control, dynamic route management, and zero-downtime updates.

Personal Projects

Recruitr [Java, Spring Boot, Spring Data, Spring Security, MySQL, Kubernetes]

GitHub

- Designed a backend server to manage organizations, candidates, jobs, and vendor/client relationships.

ScoutlyMCP Server [Python, FastMCP, ADB, Claude]

GitHub

- Built an MCP server that provides Android SMS data as context to LLMs via ADB integration.

Education & Certification

ADG Institute of Technology & Management, GGSIPU

New Delhi

- Stream:** B.Tech in Computer Science and Engineering | **Performance:** 85.40% | **CGPA:** 9.13

Integrated Institute of Technology, BTE

New Delhi

- Stream:** Diploma in Information Technology ESM | **Performance:** 70%

Certifications: Spring Boot 3 & Spring Framework 6 | Microservices with Spring Cloud

Udemy